

Calculus 2 - Homework 2 (Worked Solutions)

1.

Repeated linear factor $(x-1)^2$ needs two terms:

$$1/(x(x-1)^2) = A/x + B/(x-1) + C/(x-1)^2.$$

$$1 = A(x-1)^2 + Bx(x-1) + Cx.$$

$$x = 0: A = 1. \quad x = 1: C = 1.$$

$$\text{Coefficient of } x^2: A + B = 0 \rightarrow B = -1.$$

$$\text{integral } [1/x - 1/(x-1) + 1/(x-1)^2] dx$$

$$= \ln|x| - \ln|x-1| - 1/(x-1) + C.$$

$$\text{ANSWER: } \ln|x/(x-1)| - 1/(x-1) + C$$

2.

This is a trigonometric integral in powers of tan and sec. Since $\sec^2 x$ is the derivative of $\tan x$, the natural handle is $u = \tan x$.

$$\text{Let } u = \tan x, \quad du = \sec^2 x \, dx.$$

$$\text{integral } \tan^2 x \sec^2 x \, dx = \text{integral } u^2 \, du = u^3/3 + C.$$

$$\text{Back-substitute: } \tan^3 x / 3 + C.$$

$$\text{ANSWER: } \tan^3(x)/3 + C$$

3.

$$\text{Substitute } x = 3 \sin \theta, \quad dx = 3 \cos \theta \, d\theta,$$

$$\sqrt{9 - x^2} = 3 \cos \theta.$$

$$\text{integral } 3 \cos \theta * 3 \cos \theta \, d\theta = 9 \text{ integral } \cos^2 \theta \, d\theta.$$

$$\text{Power-reduce: } 9 \text{ integral } (1 + \cos 2\theta)/2 \, d\theta = (9/2)(\theta + \sin(2\theta)/2).$$

$$= (9/2) \theta + (9/2)(\sin \theta \cos \theta) \text{ since } \sin 2\theta = 2 \sin \theta \cos \theta.$$

$$\text{Now } \sin \theta = x/3 \text{ and } \cos \theta = \sqrt{9 - x^2}/3, \text{ so}$$

$$\sin \theta \cos \theta = x \sqrt{9 - x^2}/9, \text{ and } \theta = \arcsin(x/3).$$

$$\text{Result} = (9/2) \arcsin(x/3) + (9/2)(x \sqrt{9 - x^2}/9)$$

$$= (9/2) \arcsin(x/3) + (x/2) \sqrt{9 - x^2} + C.$$

$$\text{ANSWER: } (x/2) \sqrt{9 - x^2} + (9/2) \arcsin(x/3) + C$$

4.

This looks like a trigonometric-integral reduction, but it is a clean substitution because $\sec^2(x)$ is the derivative of $\tan(x)$.

$$\text{Let } u = \tan(x), \text{ so } du = \sec^2(x) \, dx.$$

$$\text{Change limits: } x = 0 \text{ gives } u = \tan 0 = 0; \quad x = \pi/4 \text{ gives } u = \tan(\pi/4) = 1.$$

$$\text{The integral becomes } \text{integral}_{0^1} u \, du = [u^2/2]_{0^1} = 1/2 - 0 = 1/2.$$

$$\text{ANSWER: } 1/2$$

5.

This is a cyclic parts problem. Let $I = \text{integral } e^x \sin x \, dx$.

$$\text{Parts once: } u = \sin x, \quad dv = e^x \, dx \rightarrow du = \cos x \, dx, \quad v = e^x.$$

$$I = e^x \sin x - \text{integral } e^x \cos x \, dx.$$

$$\text{Parts again on integral } e^x \cos x \, dx: \quad u = \cos x, \quad dv = e^x \, dx \rightarrow$$

$$\text{integral } e^x \cos x \, dx = e^x \cos x - \text{integral } e^x(-\sin x) \, dx$$

$$= e^x \cos x + \text{integral } e^x \sin x \, dx = e^x \cos x + I.$$

$$\text{Substitute back: } I = e^x \sin x - (e^x \cos x + I) = e^x \sin x - e^x \cos x - I.$$

$$\text{So } 2I = e^x \sin x - e^x \cos x, \text{ giving } I = (e^x (\sin x - \cos x))/2 + C.$$

$$\text{ANSWER: } (e^x (\sin(x) - \cos(x)))/2 + C$$

6.

arctan is an inverse-trig function, but this is not an integral that PRODUCES an arctan; the arctan is the integrand, so use parts.

Let $u = \arctan x$, $dv = dx \rightarrow du = 1/(1+x^2) dx$, $v = x$.

$\int \arctan x \, dx = x \arctan x - \int x/(1+x^2) \, dx$.

For $\int x/(1+x^2) \, dx$ substitute $w = 1+x^2$: it equals $(1/2) \ln(1+x^2)$.

So the antiderivative is $x \arctan x - (1/2) \ln(1+x^2)$.

Evaluate 0 to 1: at $x=1$: $1 \cdot (\pi/4) - (1/2) \ln 2 = \pi/4 - (1/2) \ln 2$.

At $x=0$: $0 - (1/2) \ln 1 = 0$.

ANSWER: $\pi/4 - (1/2) \ln(2)$

7.

Decompose $(x+3)/((x-1)(x+2)) = A/(x-1) + B/(x+2)$.

$x + 3 = A(x+2) + B(x-1)$.

$x = 1$: $4 = 3A \rightarrow A = 4/3$. $x = -2$: $1 = -3B \rightarrow B = -1/3$.

$\int [(4/3)/(x-1) - (1/3)/(x+2)] \, dx$

$= (4/3) \ln|x-1| - (1/3) \ln|x+2| + C$.

ANSWER: $(4/3) \ln|x-1| - (1/3) \ln|x+2| + C$

8.

$\int_1^\infty 1/x \, dx = \lim_{t \rightarrow \infty} [\ln x]_1^t = \lim_{t \rightarrow \infty} (\ln t - 0)$.

As $t \rightarrow \infty$, $\ln t \rightarrow \infty$, so the limit does not exist (is infinite).

This is the borderline p-integral with $p = 1$, which diverges.

ANSWER: diverges

9.

The whole term is raised to the n-th power, so the Root Test is natural.

$a_n = (n/(2n+1))^n$. Take the n-th root:

$(a_n)^{1/n} = n/(2n+1)$.

$L = \lim_{n \rightarrow \infty} n/(2n+1) = 1/2$.

Since $L = 1/2 < 1$, the series converges by the Root Test.

ANSWER: converges (root test, $L = 1/2 < 1$)

10.

Here $f(x) = 1/x$, $f'(x) = -1/x^2$. On $[1, 2]$, $|f'|$ is largest at $x = 1$,

so take $K = \max|f'| = 1$. Also $b - a = 1$.

Bound: $|E_T| \leq K(b-a)^2/(2n) = 1/(2n) = 1/(2n)$.

Require $1/(2n) < 0.001$, i.e. $2n > 1000$, so $n > 500$, $n > 12.91$.

The smallest integer is $n = 500$.

ANSWER: $n = 500$